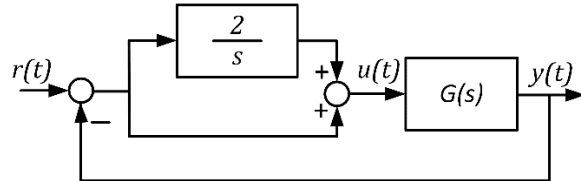


A**Name and group:** _____

This exam is closed books. Write your name on every page. Write clearly and legibly. Explain your work in words.
Good luck!

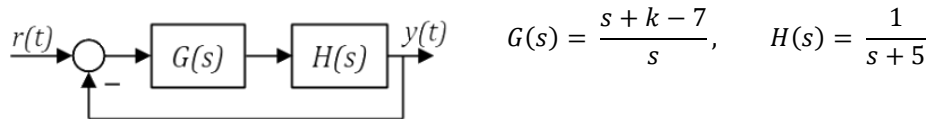
P1 (1.5p). Consider the closed-loop system shown in the figure:

A) (0.5p) Determine the transfer function $G(s)$ if the differential equation that describes the relationship between the input $u(t)$ and the output $y(t)$ of this block is:

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 2y(t) = \frac{du(t)}{dt}$$

B) (0.5p) Compute the equivalent closed-loop transfer function from the input $r(t)$ to the output $y(t)$.

C) (0.5p) Determine the steady-state error for a unit step input $r(t) = 1, t \geq 0$.

P2 (2.5p). Consider a unity feedback closed-loop system, where $k \in [0, \infty)$ and:

$$G(s) = \frac{s+k-7}{s}, \quad H(s) = \frac{1}{s+5}$$

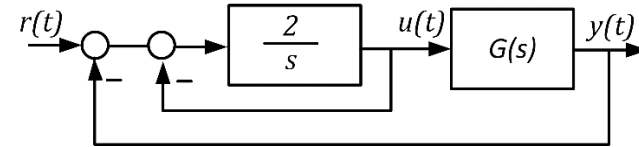
A) (1.5p) Sketch the root locus for the parameter $k \in [0, \infty)$.

B) (0.5p) Determine the values of the parameter k for which the system is stable and describe the result on the root locus.

C) (0.5p) Determine the values of the parameter k for which the system is overdamped and describe the result on the root locus.

B**Name and group:** _____

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Good luck!

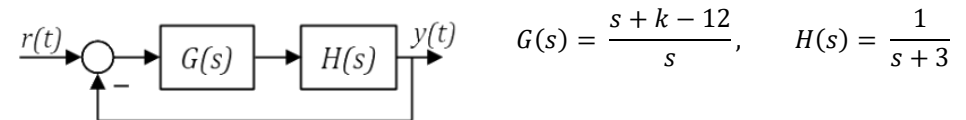
P1 (1.5p). Consider the closed-loop system shown in the figure:

A) (0.5p) Determine the transfer function $G(s)$ if the differential equation that describes the relationship between the input $u(t)$ and the output $y(t)$ of this block is:

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + 2y(t) = \frac{du(t)}{dt} + 2u(t)$$

B) (0.5p) Compute the equivalent closed-loop transfer function from the input $r(t)$ to the output $y(t)$.

C) (0.5p) Determine the steady-state error for a unit step input $r(t) = 1, t \geq 0$.

P2 (2.5p). Consider a unity feedback closed-loop system, where $k \in [0, \infty)$ and:

$$G(s) = \frac{s+k-12}{s}, \quad H(s) = \frac{1}{s+3}$$

A) (1.5p) Sketch the root locus for the parameter $k \in [0, \infty)$.

B) (0.5p) Determine the values of the parameter k for which the system is stable and describe the result on the root locus.

C) (0.5p) Determine the values of the parameter k for which the system is underdamped and describe the result on the root locus.

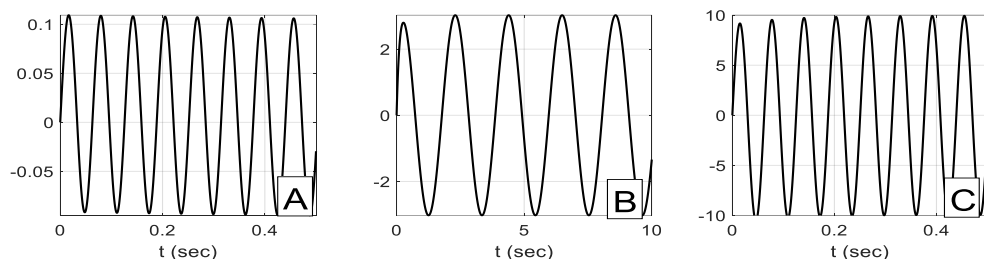
Name and group: _____

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P1 (2p). For a system with a transfer function: $G(s) = \frac{10(s+1)}{s+10}$

A) (1p) Draw the Bode diagram.

B) (1p) Two of the figures below show the output signals of this system for the inputs $u_1(t) = \sin(100t)$ and $u_2(t) = \sin(3t)$. Choose the appropriate figures, clearly motivate why you chose them and why the third figure cannot be the correct answer.



P2 (2p). A unity negative feedback control system has an open loop transfer function:

$$G(s) = \frac{1}{s(s+1)(s+4)}$$

A) (1.5p) Design a PD compensator such that the dominant complex closed-loop poles have a damping ratio $\zeta = 0.5$ and the natural frequency $\omega_n = 2$ rad/s.

B) (0.5p) Consider the unity negative feedback control system with the plant $G(s)$ and the compensator designed at A). Can you improve the steady-state error for a ramp input? If the answer is yes, what type of controller would you use and why? If the answer is no, motivate your response.

P3. (1p) Consider a discrete system with the transfer function:

$$G(z) = \frac{z}{10z^2 + (10a+1)z + a}$$

Analyze the discrete system stability for all values of the real parameter a .

P4 (1p). A system with the input u and the output y has a transfer function: $G(s) = \frac{2}{s+2}$

A) (0.5p) Choose a sampling period (T_e), motivate your choice and determine the discrete transfer function $G(z)$ using the transformation $s = \frac{1-z^{-1}}{T}$.

B) (0.5p) Determine the difference equation for the discrete system.

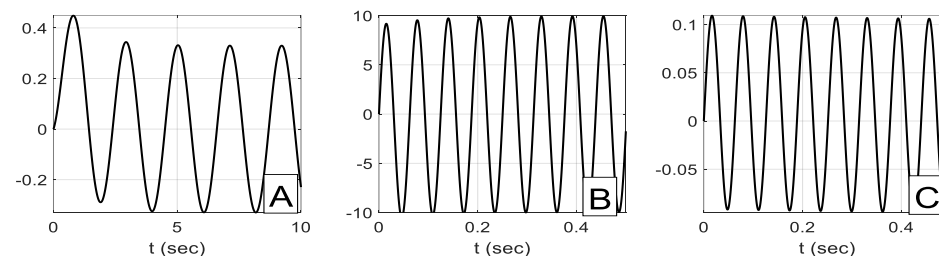
Name and group: _____

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P1 (2p). For a system with a transfer function: $G(s) = \frac{0.1(s+10)}{s+1}$

A) (1p) Draw the Bode diagram.

B) (1p) Two of the figures below show the output signals of this system for the inputs $u_1(t) = \sin(100t)$ and $u_2(t) = \sin(3t)$. Choose the appropriate figures, clearly motivate why you chose them and why the third figure cannot be the correct answer.



P2 (2p). A unity negative feedback control system has an open loop transfer function:

$$G(s) = \frac{1}{(s+1)(s+4)}$$

A) (1.5p) Design a PI compensator such that the dominant complex closed-loop poles have a damping ratio $\zeta = 0.5$ and the natural frequency $\omega_n = 2$ rad/s.

B) (0.5p) Consider the unity negative feedback control system with the plant $G(s)$ and the compensator designed at A). Can you improve the steady-state error for a step input? If the answer is yes, what type of controller would you use and why? If the answer is no, motivate your response.

P3. (1p) Consider a discrete system with the transfer function:

$$G(z) = \frac{z-2}{z^2 + (a+10)z + 10a}$$

Analyze the discrete system stability for all values of the real parameter a .

P4 (1p). A system with the input u and the output y has a transfer function: $G(s) = \frac{2}{2s+1}$

A) (0.5p) Choose a sampling period (T_e), motivate your choice and determine the discrete transfer function $G(z)$ using the transformation $s = \frac{1-z^{-1}}{T}$.

B) (0.5p) Determine the difference equation for the discrete system.